

Passing the Part 107 Test

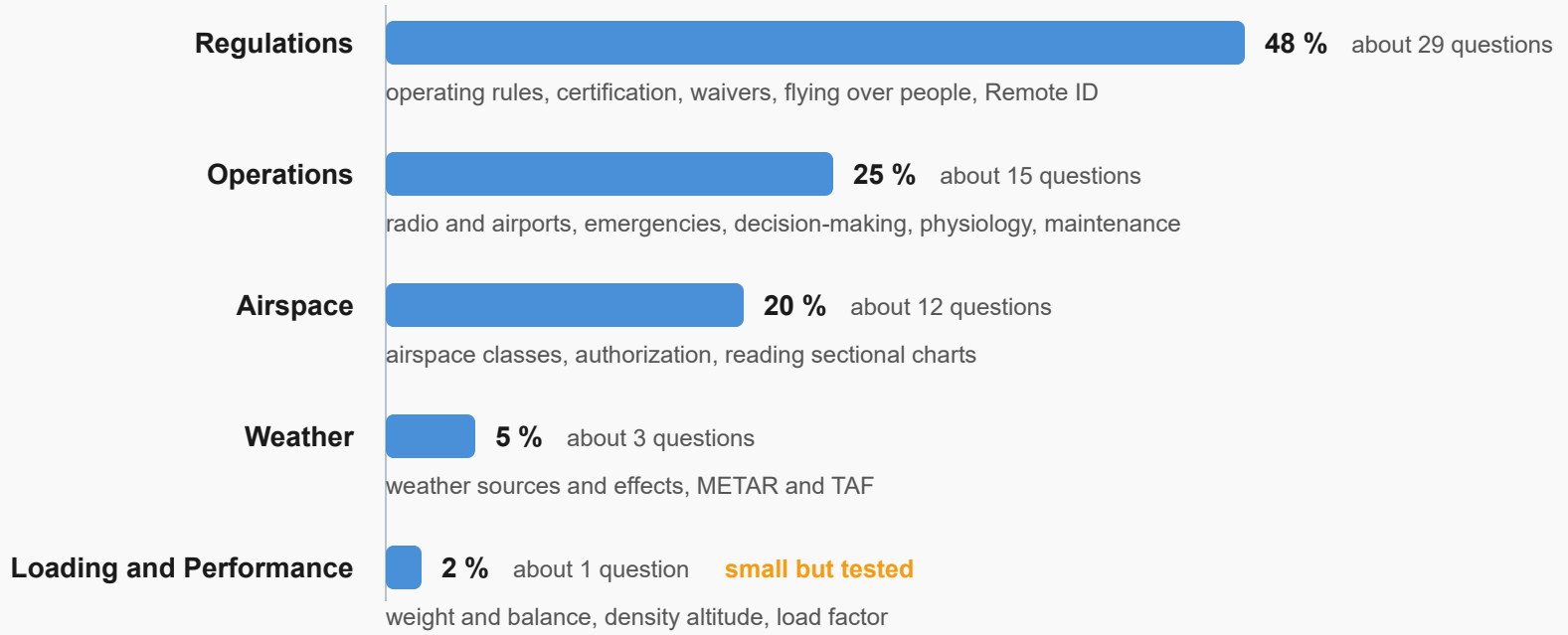
Week 9 — Tuesday Lecture

The test

Questions	60 scored + 5 unscored, multiple choice A/B/C
Time	2 hours
Pass	70 % (42 of 60)
Where	PSI testing centres — locally, UVU's centre at the Provo airport
Bring	Photo ID with date of birth, signature and address
Provided	Calculator built in; the figures (on screen or the supplement book)
Retake	After 14 days, with your failed test report
Fee	*As of October 2026, about $175 — check PSI when you book*

Figure 7: What the knowledge test covers

60 scored questions. Weights from the PSI applicant bulletin, effective 29 Sept 2025; older study guides show different ranges.



Regulations alone are nearly half the test. Five unscored trial questions are mixed in, so the screen shows 65.

Half the test is regulations. Weather and loading are small, but they are on it.

How to read a figure question

"(Refer to FAA-CT-8080-2H, Figure 26, area 2.)"

1. Find **Figure 26** in the testing supplement — or on screen, inside the question
2. Find the **red number 2** on the chart
3. Read the question about *that* spot
4. When a symbol is unfamiliar, **use the legend** — it is in the supplement too

Rapid round — seen in Week 3

1. Maximum altitude away from structures? A) 400 ft AGL · B) 400 ft MSL · C) 500 ft AGL
2. Minimum visibility? A) 1 SM · B) 3 SM · C) 5 SM
3. Flying 20 minutes after sunset needs? A) a waiver · B) anti-collision lights · C) nothing
4. Hours from last drink to flying? A) 4 · B) 8 · C) 12
5. Which Part 107 aircraft must be registered? A) over 0.55 lb · B) all of them · C) over 55 lb
6. Damage that must be reported? A) any · B) over $500 · C) over $1,000

ANSWER **A · B · B · B · B · B**

1. **400 ft AGL** — or 400 ft above a structure you are within 400 ft of
2. **3 statute miles** from the control station
3. **Anti-collision lights** visible 3 SM — night flying is allowed since 2021
4. **8 hours**, and below 0.04 % blood alcohol
5. **All of them** — 0.55 lb is the *recreational* threshold
6. **Over $500**, not counting the drone itself — report to the FAA within 10 days

Who issues a Declaration of Compliance?

A) The remote pilot · B) The aircraft's manufacturer, accepted by the FAA · C) The owner, after registering

ANSWER B — the manufacturer declares, the FAA accepts

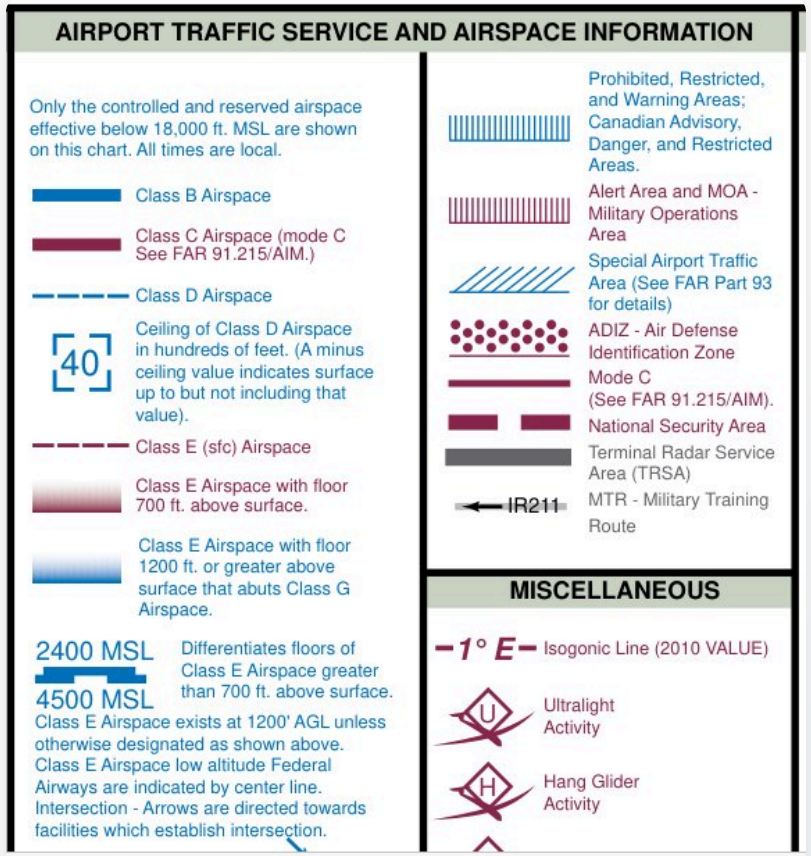
- Category 2 and 3 aircraft are on an **FAA-accepted Declaration of Compliance** from the manufacturer
- The pilot's job is to **check the category on the label** before flying over people
- Category 1: 0.55 lb or less, no exposed parts that could cut skin — no declaration needed
- Category 4: an airworthiness certificate
- Figure 15b in the reading has all four categories on one page

You climb above 600 ft AGL to avoid a manned aircraft.

To whom must you report the deviation? A) Air traffic control · B) The NTSB · C) The FAA, upon request

ANSWER C — the FAA, upon request

- In an emergency you may deviate from any rule to the extent needed (§107.21)
- You report it **only if the FAA asks**
- Accidents are different: those go to the FAA **within 10 days**, whether asked or not



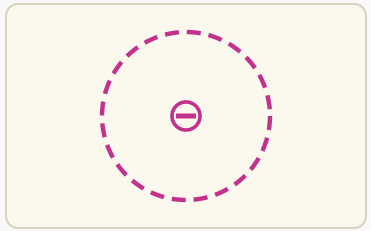
FAA-CT-8080-2H, Legend 1. Public domain. Not for navigation.

The sectional legend's airspace symbols (FAA-CT-8080-2H). When a chart symbol is unfamiliar, this is where the answer is.

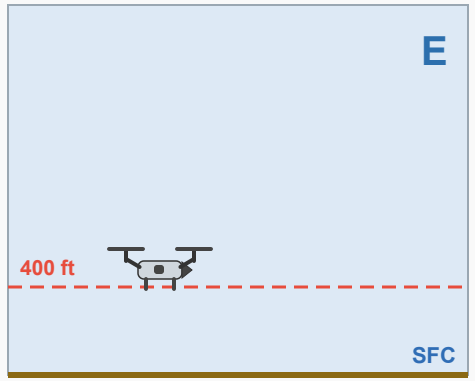
Figure 11b: Where Class E starts

Three ways the sectional draws the floor of Class E, and what each one means for a drone at or below 400 ft.

Class E from the surface



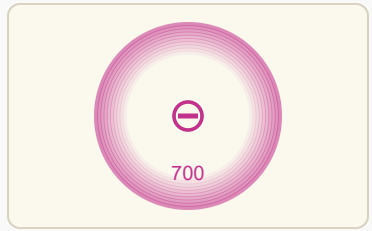
dashed magenta line



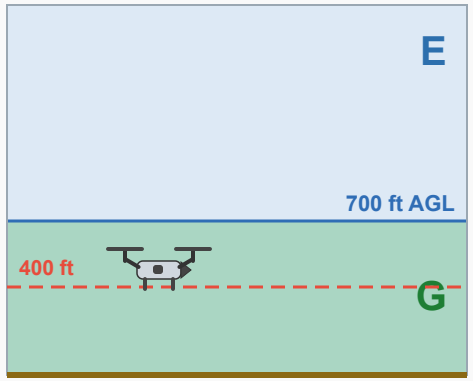
Authorization needed

Controlled from the ground up. Ask ATC first: LAANC or FAA DroneZone (14 CFR 107.41).

Class E from 700 ft AGL



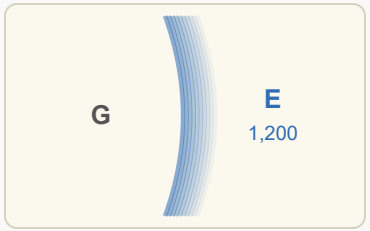
magenta vignette (shaded band)



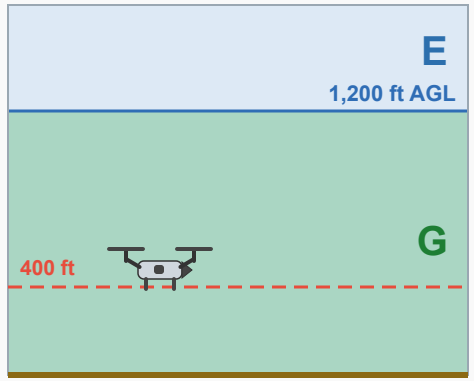
No authorization

Class G below 700 ft. Your 400 ft is uncontrolled airspace: fly, no permission needed.

Class E from 1,200 ft AGL



blue vignette, or no line at all



No authorization

The default over most of the US. Class G up to 1,200 ft, so the same answer: fly.

Only surface Class E needs authorization for a drone below 400 ft. Symbols redrawn from the chart legend, not to scale.

Class E can start at the surface, 700 ft or 1,200 ft. Only surface Class E needs authorization below 400 ft.

Savannah, area 3: what is the floor of the Class C shelf (outer circle)?

A) 1,300 ft AGL · B) 1,300 ft MSL · C) 1,700 ft MSL



FAA-CT-8080-2H, Figure 23. Public domain. Not for navigation.

ANSWER **B — 1,300 ft MSL**

- "41/13" means **ceiling 4,100 ft, floor 1,300 ft**, in hundreds of feet
- Airspace altitudes on a sectional are **MSL** — above sea level
- The 400 ft drone limit is **AGL** — above the ground
- Subtract the ground elevation to compare the two

Inspecting a tower near Jamestown Regional, area 4. What do you need first?

A) Military authorization · B) ATC authorization · C) National Park Service authorization



FAA-CT-8080-2H, Figure 26. Public domain. Not for navigation.

ANSWER B — it is surface Class E

- The **dashed magenta line** around Jamestown marks Class E that starts at the **surface**
- Surface Class E is controlled: you need **ATC authorization** (§107.41), usually through LAANC
- A **magenta** shaded edge means Class E from 700 ft; **blue** means 1,200 ft — no authorization needed below 400 ft

Area 2: the gray line labelled "VR1667, VR1617…". Is it a hazard?

A) No, all operations are above 400 ft · B) Yes, a military training route from the surface to 1,500 ft AGL · C) Yes, the route separates traffic



FAA-CT-8080-2H, Figure 59. Public domain. Not for navigation.

ANSWER **B — a military training route, surface to 1,500 ft AGL**

- **VR** = flown under visual flight rules; **IR** = instrument rules
- **Four digits**: at or below 1,500 ft AGL. **Three digits**: some segments above 1,500 ft
- Military jets fly these fast and low — be alert near them, though they are not off-limits

A crop survey in the Devils Lake West MOA. How do you find out whether it is active?

A) The chart legend · B) A small UAS database · C) A military operations directory



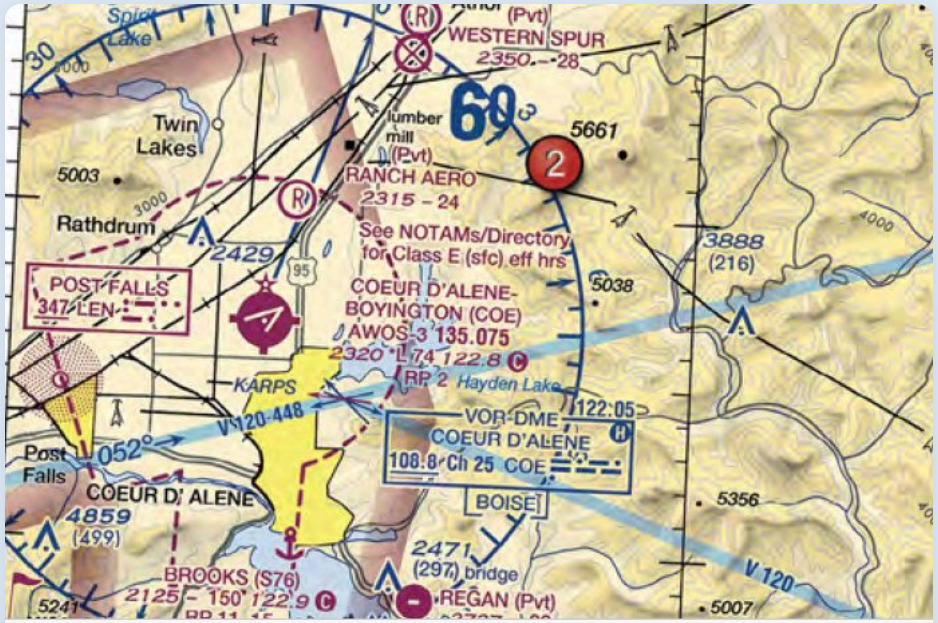
FAA-CT-8080-2H, Figure 21. Public domain. Not for navigation.

ANSWER **A — the chart legend**

- A full sectional lists each special-use area's times of use in its legend and margin tables
- A **military operations area (MOA)** is a hazard to know about, not a ban
- A **restricted area** needs the controlling agency's permission
- A **TFR** comes by NOTAM — check before every flight

Area 2: what is the CTAF at Coeur d'Alene?

A) 122.05 · B) 135.075 · C) 122.8



FAA-CT-8080-2H, Figure 22. Public domain. Not for navigation.

ANSWER **C — 122.8**

- **CTAF**: the common traffic advisory frequency, where pilots announce their positions at an airport without an operating tower
- On the chart it is the frequency with a **C in a circle** after it
- A drone pilot doesn't have to transmit, but **listening** tells you where manned traffic is

Figure 11c: Reading a traffic pattern

Runway 13, standard left-hand pattern, north up. Where is an aircraft that reports “midfield left downwind, runway 13”?

How to read it

Runway number = heading ÷ 10

Runway 13 lands toward 130° magnetic. Runway 31 is the same strip from the other end, heading 310°.

Standard pattern: all turns left

Unless the chart says RP, right pattern.

Downwind: parallel, opposite way

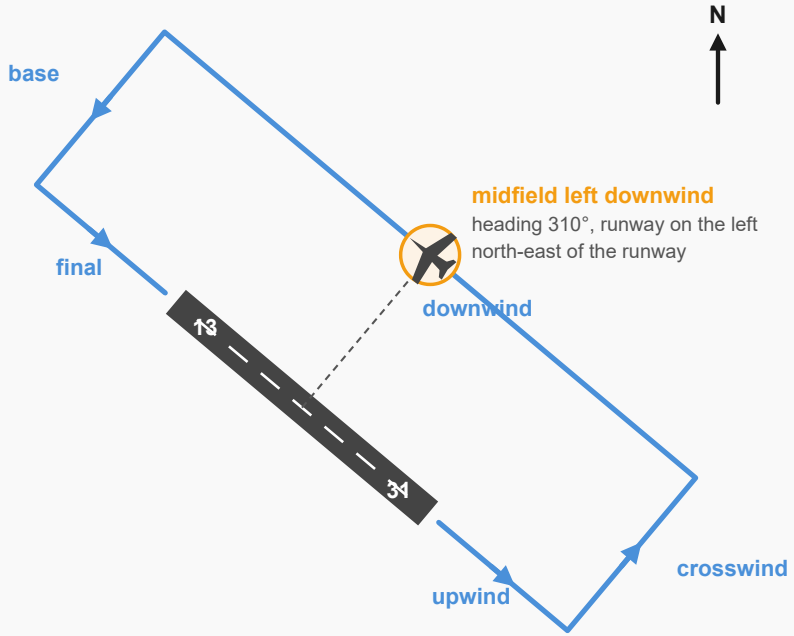
The runway stays on the pilot’s left, so the downwind of runway 13 is flown on 310° along the north-east side.

The test answer

“Midfield left downwind, runway 13” puts the aircraft east of the runway (FAA sample question, Figure 26, area 2).

Why a drone pilot cares

Stay out of the pattern (14 CFR 107.43). Hear a position report, know where to look.



Upwind, crosswind, downwind, base, final: five legs, four left turns. Draw the pattern from the runway number and the answer falls out.

Runway 13 points about 130° magnetic. A left pattern turns left; downwind runs parallel, the opposite way.

On the Cooperstown CTAF: "midfield left downwind, runway 13."

Where is that aircraft? A) East of the runway · B) South of the runway · C) West of the runway



FAA-CT-8080-2H, Figure 26. Public domain. Not for navigation.

ANSWER A — east of the runway

- Runway 13: landing toward about **130°** magnetic
- Left traffic: all turns are to the **left**, so the pattern sits on the **left** of the landing direction
- Downwind runs the opposite way, about 310°, on that side — **northeast, to the east** of the runway
- Draw it: runway, landing arrow, turn left four times

Figure 24: Five hazardous attitudes and their antidotes

Recognise the thought, name the attitude, say the antidote. Wording from the FAA Pilot's Handbook, chapter 2.

	The attitude, and what it sounds like	The antidote: say it to yourself
1	**Anti-authority** *"Don't tell me."*	**Follow the rules. They are usually right.**
2	**Impulsivity** *"Do it quickly."*	**Not so fast. Think first.**
3	**Invulnerability** *"It won't happen to me."*	**It could happen to me.**
4	**Macho** *"I can do it."*	**Taking chances is foolish.**
5	**Resignation** *"What's the use?"*	**I'm not helpless. I can make a difference.**

The test tells a short story and asks which attitude it shows. "Fly first, ask questions later" is impulsivity; "watch this" is macho; "the rules are for other people" is anti-authority.

Five hazardous attitudes, and what to tell yourself instead.

Your station manager says "fly first, ask questions later." Which hazardous attitude?

A) Macho · B) Invulnerability · C) Impulsivity

ANSWER C — impulsivity

- Impulsivity: "Do it quickly." Antidote: **"Not so fast. Think first."**
- Macho is "I can do it"; invulnerability is "It won't happen to me"
- On the test, match the **words** to the attitude

Figure 23 — IMSAFE Personal Fitness Checklist

I - M - S - A - F - E

I	**Illness**	Am I suffering from any illness or symptom of illness? Do not fly if sick.
M	**Medication**	Am I taking any medications that could impair judgment or reaction time?
S	**Stress**	Am I under psychological pressure? Stress narrows attention and impairs decisions.
A	**Alcohol**	8 hours bottle-to-throttle. No flying under the influence (0.04% BAC limit per FAR).
F	**Fatigue**	Am I adequately rested? Fatigue degrades performance as much as alcohol.
E	**Emotion / Eating**	Am I emotionally upset or hungry/dehydrated? Strong emotions impair judgment.

If the answer to ANY of these is "yes," consider postponing the flight.

IMSAFE: check yourself before you check the aircraft.

A battery is swollen and warm before a flight.

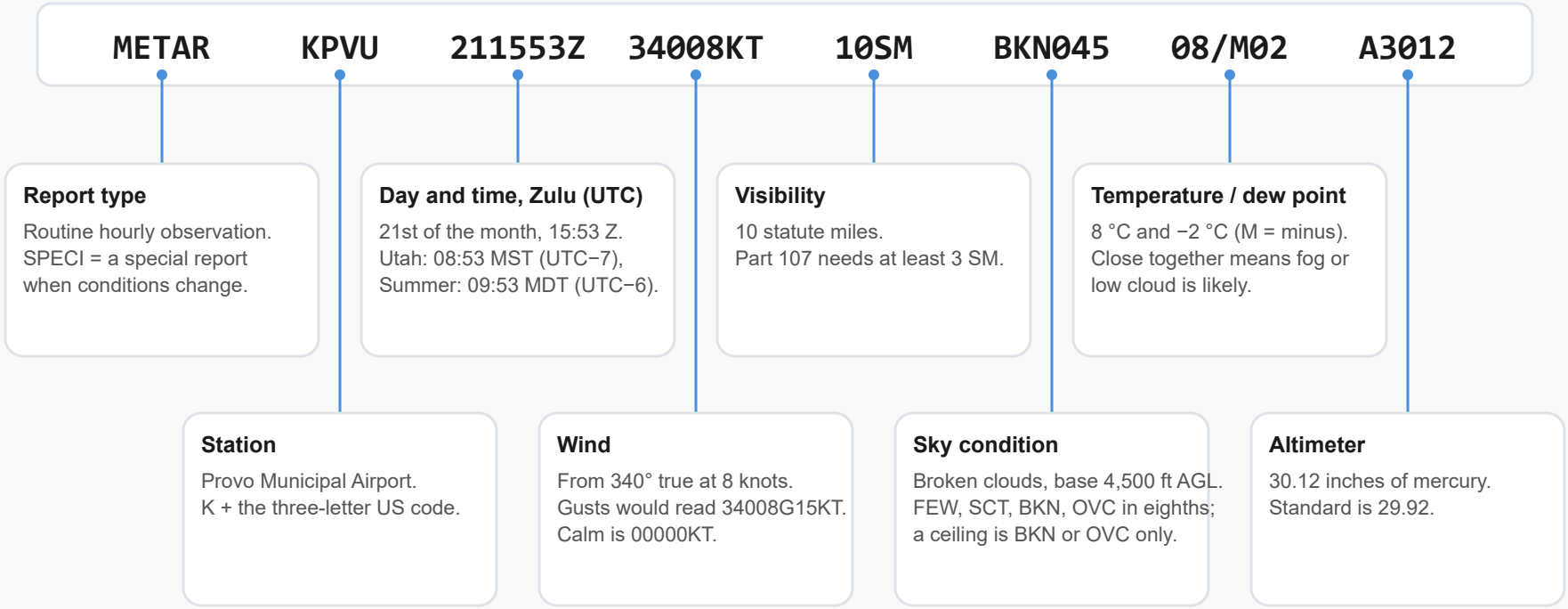
What do you do? A) Fly a short flight · B) Cool it in the fridge · C) Don't fly it; move it outdoors onto a non-flammable surface

ANSWER C — never fly a damaged lithium battery

- Swelling, heat or smell means the cell is damaged and can catch fire
- Move it **outdoors**, onto sand or concrete, away from people and anything that burns
- Follow the manufacturer's disposal instructions
- Lithium fires are hard to put out — keep a fire bag or sand bucket in the kit

Figure 16: Reading a METAR

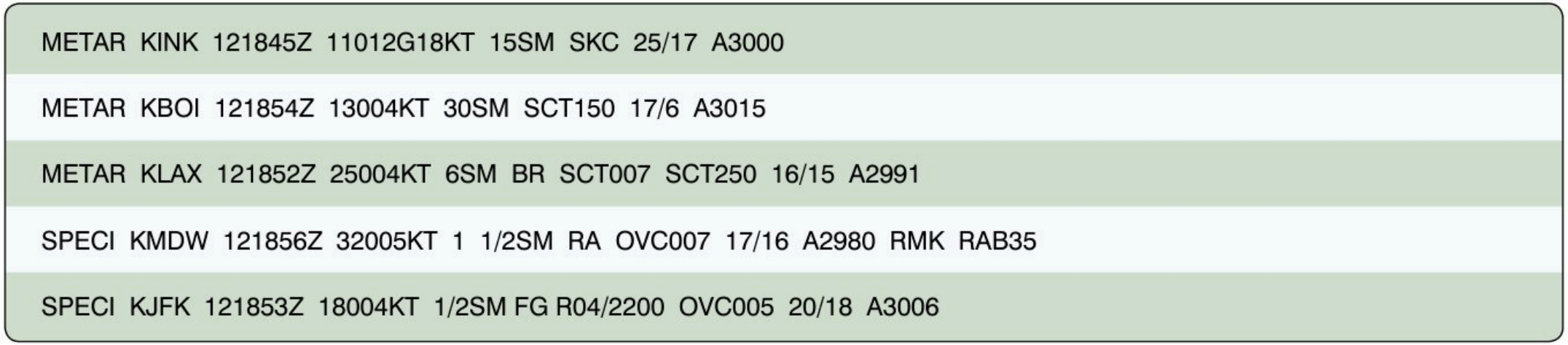
An example in the real format, not an actual observation: a winter morning at Provo. Read it group by group.



A METAR, decoded. Weather is about 5 % of the test.

KMDW: "32005KT 1 1/2SM RA OVC007". What are the conditions?

A) 700 ft overcast, visibility 1½ SM, rain · B) 7,000 ft overcast, 1½ SM, heavy rain · C) 700 ft overcast, visibility 11, occasionally 2 SM, rain



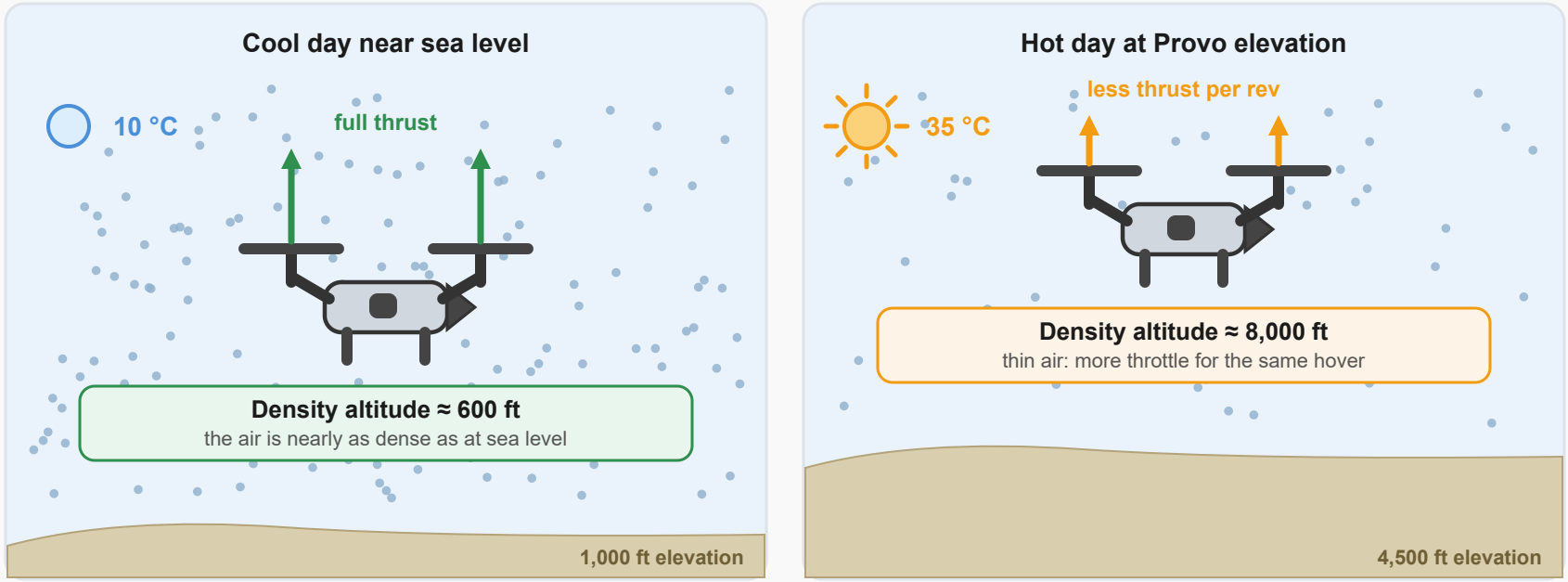
FAA-CT-8080-2H, Figure 12. Public domain. Not for navigation.

ANSWER A — overcast at 700 ft AGL, 1½ statute miles, rain

- Cloud height in **hundreds of feet AGL**: 007 = 700 ft
- A **ceiling** is the lowest **broken (BKN)** or **overcast (OVC)** layer
- Visibility below 3 SM: **no Part 107 flight**
- **RA** alone is moderate rain; **+RA** would be heavy
- Watch for: times are **Zulu** (Utah is UTC−7 in winter, −6 in summer); winds are **true**

Figure 19: Hot and high: density altitude

The same aircraft on two days. Thin air gives the propellers less to push on, and the aircraft flies as if it were much higher than it is.



High density altitude reduces propeller efficiency and performance.

Less thrust and less lift: a weaker climb, more throttle to hover, a shorter flight, and a longer takeoff for a fixed wing.

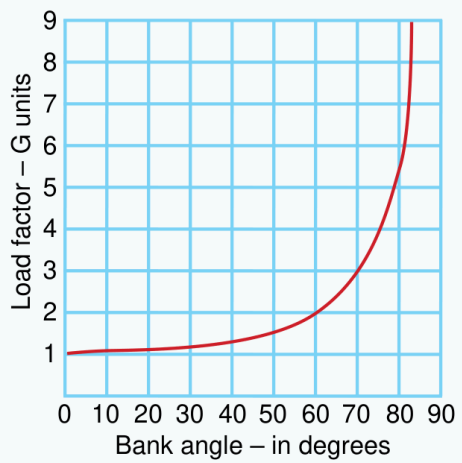
Density altitude rises with elevation, temperature and humidity. Rule of thumb: about 120 ft per °C above the standard day (15 °C at sea level, 2 °C cooler per 1,000 ft). Pilot's Handbook of Aeronautical Knowledge, chapter 11.

Hot and high: thinner air means less lift. Loading and performance is about 2 % of the test.

A 33 lb unmanned airplane in a 30° level banked turn.

What load must its structure support? A) 34 lb · B) 47 lb · C) 38 lb

Angle of bank Φ	Load factor n
0°	1.0
10°	1.015
30°	1.154
45°	1.414
60°	2.000
70°	2.923
80°	5.747
85°	11.473
90°	∞



FAA-CT-8080-2H, Figure 2. Public domain. Not for navigation.

ANSWER C — about 38 lb

- Load factor in a 30° bank is **1.154**, read from the chart (it is 1 ÷ cos 30°)
- 33 lb × 1.154 ≈ **38 lb**
- 45° → 1.41; 60° → 2.0 — steep turns get expensive fast
- Figure 20b in the reading draws all four bank angles

The traps

- Chart altitudes are **MSL**; cloud heights and the 400 ft limit are **AGL**
- A **ceiling** is BKN or OVC only
- METAR times are **Zulu**; METAR winds are **true**
- **Surface** Class E needs authorization; 700 ft and 1,200 ft Class E don't, below 400 ft
- MOA = hazard; restricted = permission; TFR = by NOTAM
- Emergency deviations: report to the FAA **on request**; accidents: **within 10 days**

A two-week study plan

- **Week 1:** the overview reading and the knowledge review; FAA sample questions; AC 107-2A
- **Week 2:** charts every day — the supplement figures and the legend; the practice exams here
- **Day before:** the traps list; sleep
- Free: FAA sample questions · AC 107-2A · the Pilot's Handbook of Aeronautical Knowledge · PSI's sample test
- Knowledge Review · Study Resources

What to do Thursday

- **Part 107 study lab:** Part 107 Studying
- The **practice exam** opens this week — see Learning Suite
- Practice: Mini Practice Exam · Full Practice Exam